

Historical Air Quality Analysis

Air Quality UCI dataset · Italian city · March 2004 – April 2005

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Executive Summary

This report analyses 9,357 hourly air-quality and meteorological observations from the UCI Air Quality dataset, focusing on CO, benzene (C₆H₆), NO_x, NO₂, temperature, relative humidity and absolute humidity, collected by a multi-sensor field device in an Italian city between 10 March 2004 and 04 April 2005. The objective is to identify temporal patterns, quantify pollutant relationships, and assess what these patterns imply for public health and urban policy.

Headline findings:

- **Pollutants show a pronounced diurnal pattern.** CO, benzene and NO_x show sharp morning and evening peaks, with maxima around 08:00 and 19:00 and a deep early-morning trough. Mean CO at 19:00 is roughly five times the 05:00 mean. The timing of these peaks is consistent with increased combustion-related activity during commuting hours.
- **Benzene concentrations are elevated relative to established benchmarks.** The mean concentration of 10.1 µg/m³ is approximately twice the EU calendar-year limit value of 5 µg/m³ and substantially above the 1.7 µg/m³ concentration that WHO associates with an excess lifetime leukaemia risk of 1 in 100,000.
- **Pollutants are highly correlated.** CO and benzene correlate at $r = 0.91$, and CO and NO_x at $r = 0.80$, revealing a strong combustion-related pollutant cluster consistent with shared sources such as road traffic.
- **Weekend NO_x is approximately 27% lower than weekday levels**, revealing a pronounced weekly pollution pattern consistent with reduced traffic activity at weekends. Weather variables alone are weak predictors of pollution but interact with seasonal and pollutant patterns.
- **A Random Forest model predicts hourly benzene with $R^2 \approx 0.77$ on the held-out period**, using CO, NO_x, NO₂, meteorological and temporal variables. This strong out-of-sample performance demonstrates substantial predictive signal for benzene and highlights the potential value of multivariate environmental monitoring data.

1. Dataset and Methodology

1.1 Source

Data are taken from the UCI Machine Learning Repository “Air Quality” dataset (De Vito et al., University of Salerno). Hourly averages were recorded by a multi-sensor device deployed at road level in a significantly polluted area of an Italian city.

1.2 Variables

Variable	Units	Description
CO(GT)	mg/m ³	Reference CO analyser (true value)
C6H6(GT)	µg/m ³	Reference benzene analyser
NOx(GT)	ppb	Reference NO _x analyser
NO2(GT)	µg/m ³	Reference NO ₂ analyser
PT08.S1–S5	—	Tin-oxide sensor responses (CO, NMHC, NO _x , NO ₂ , O ₃)
T, RH, AH	°C, %, —	Air temperature, relative humidity, absolute humidity

1.3 Cleaning steps

- Replaced the -200 sentinel value with NaN throughout.
- Combined Date and Time columns into a Python datetime index.
- Dropped NMHC(GT) — over 90 % missing across the period.
- Time-based interpolation was applied with a six-observation limit in each direction, while unresolved missing values were retained.
- Removed two empty trailing columns and 114 fully-blank trailing rows.

2. Temporal Trends

2.1 Daily and seasonal patterns

Figure 1 plots the daily mean of each pollutant overlaid with a 7-day rolling mean. CO and benzene show a mild seasonal dip in late summer, while NO_x and NO₂ rise sharply from October onwards. These patterns are consistent with seasonal changes in emissions and atmospheric dispersion during the colder months. The dataset ends before the 2005 summer trough is visible.

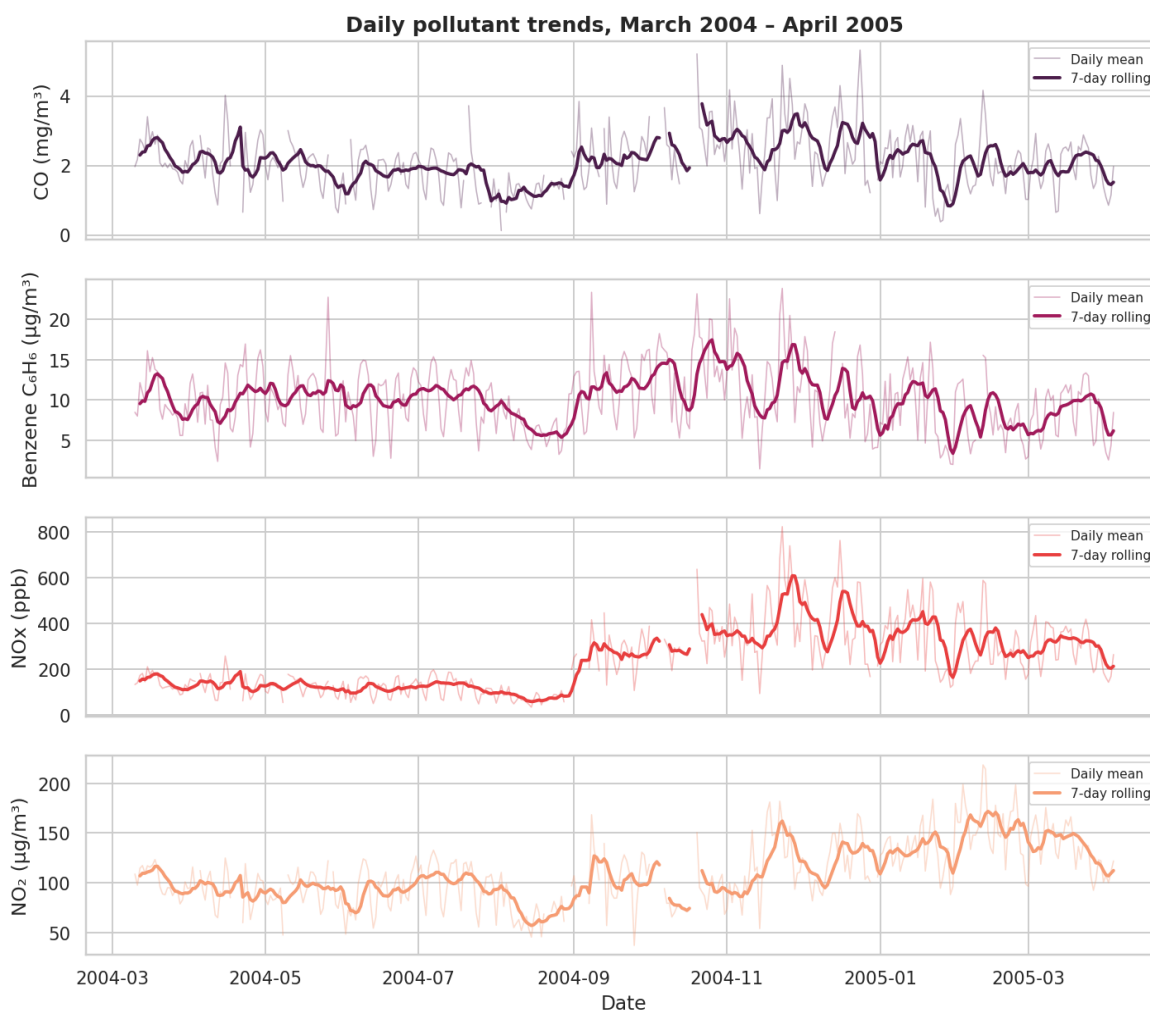


Figure 1 — Daily-mean concentrations (thin) with 7-day rolling mean (thick).

2.2 Monthly view

Normalising each pollutant to its own peak makes the seasonal structure easier to compare across species (Figure 2). All four reach near-maximum values in November–February; CO and benzene reach a clear minimum in August.

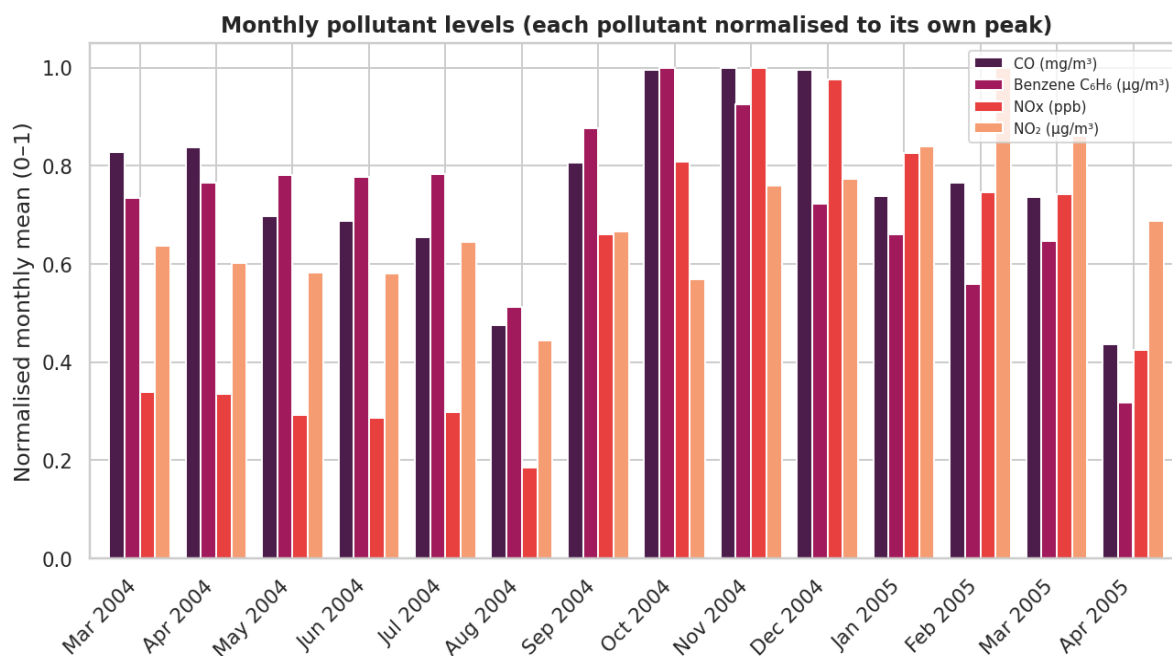


Figure 2 — Monthly means, each pollutant rescaled to its own maximum.

2.3 Diurnal cycle

Figure 3 collapses the time series onto the 24-hour clock. The twin commute peaks at 08:00 and 19:00 are unmistakable, with the evening peak the larger of the two. NO₂ remains elevated through the afternoon, giving it a broader daytime profile than the other pollutants.

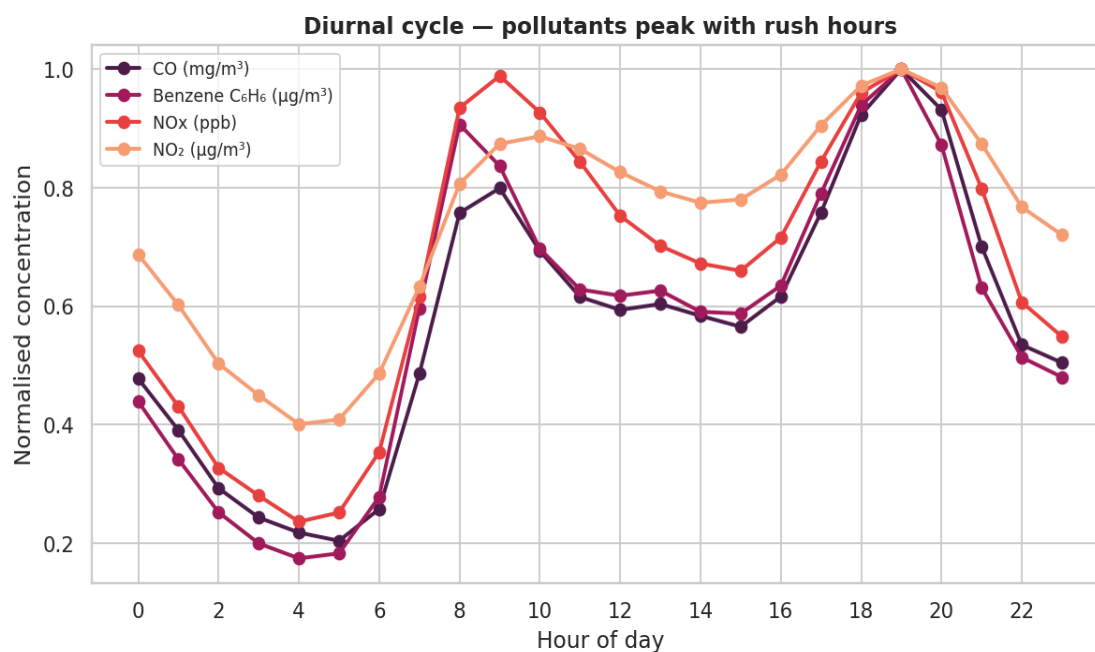


Figure 3 — Mean diurnal profile, normalised within each pollutant.

2.4 Weekday vs weekend

Box plots in Figure 4 show distributions splitting Monday–Friday from Saturday–Sunday. Median NO_x falls by about 27 % at weekends and CO by about 18 %. NO₂ shows a smaller weekend reduction than NO_x, suggesting that its weekly pattern is less pronounced than that of NO_x.

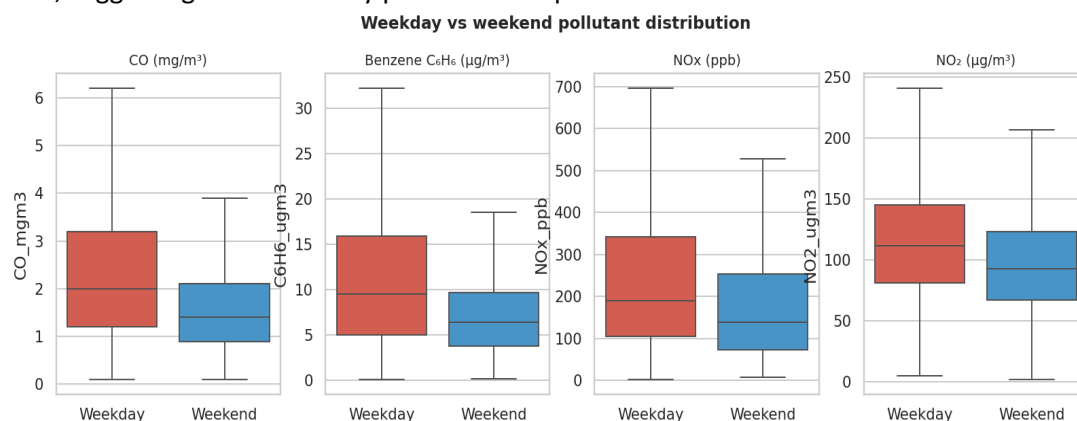


Figure 4 — Weekday vs weekend distributions (whiskers at 1.5×IQR; outliers hidden).

3. Pollutant–Weather Relationships

3.1 Correlation matrix

Figure 5 shows pairwise Pearson correlations. Three blocks stand out:

- **Combustion cluster:** CO, benzene, NO_x and NO₂ are mutually correlated ($r \approx 0.6\text{--}0.9$), revealing a strong combustion-related pollutant cluster consistent with shared sources such as road traffic.
- **Weather cluster:** Temperature and absolute humidity correlate strongly and positively ($r = 0.66$) because warm air holds more water vapour. Temperature and relative humidity correlate negatively ($r = -0.58$), the standard inverse relationship.
- **Cross terms:** NO_x and NO₂ correlate weakly but negatively with temperature ($r \approx -0.2$), reflecting better dispersion in warmer, more convective conditions; NO₂ correlates negatively with absolute humidity ($r = -0.33$).

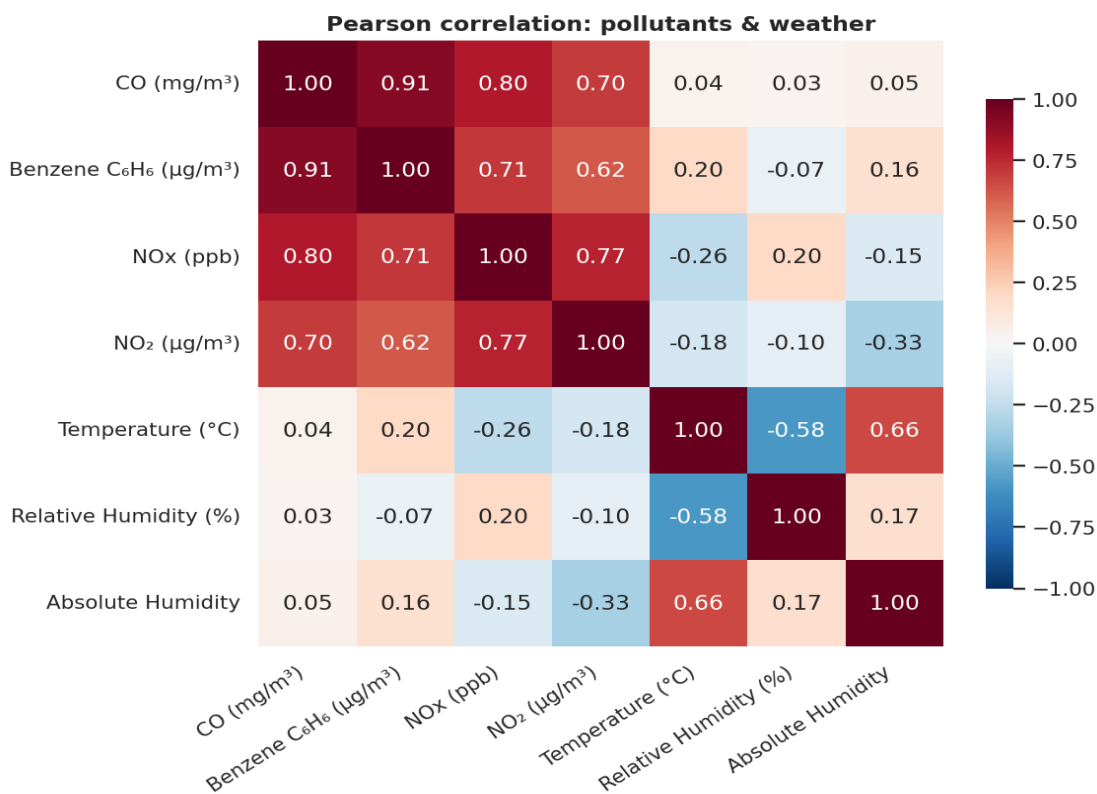


Figure 5 — Pearson correlation matrix for the four reference pollutants and three weather variables.

3.2 Scatter views

Figure 6 illustrates the cross-pollutant–weather relationships. The CO–temperature relationship is essentially flat ($r \approx 0.04$), indicating little linear association between CO concentration and temperature in this dataset. NO₂ shows a modest negative slope against relative humidity, although a substantial amount of variance remains unexplained.

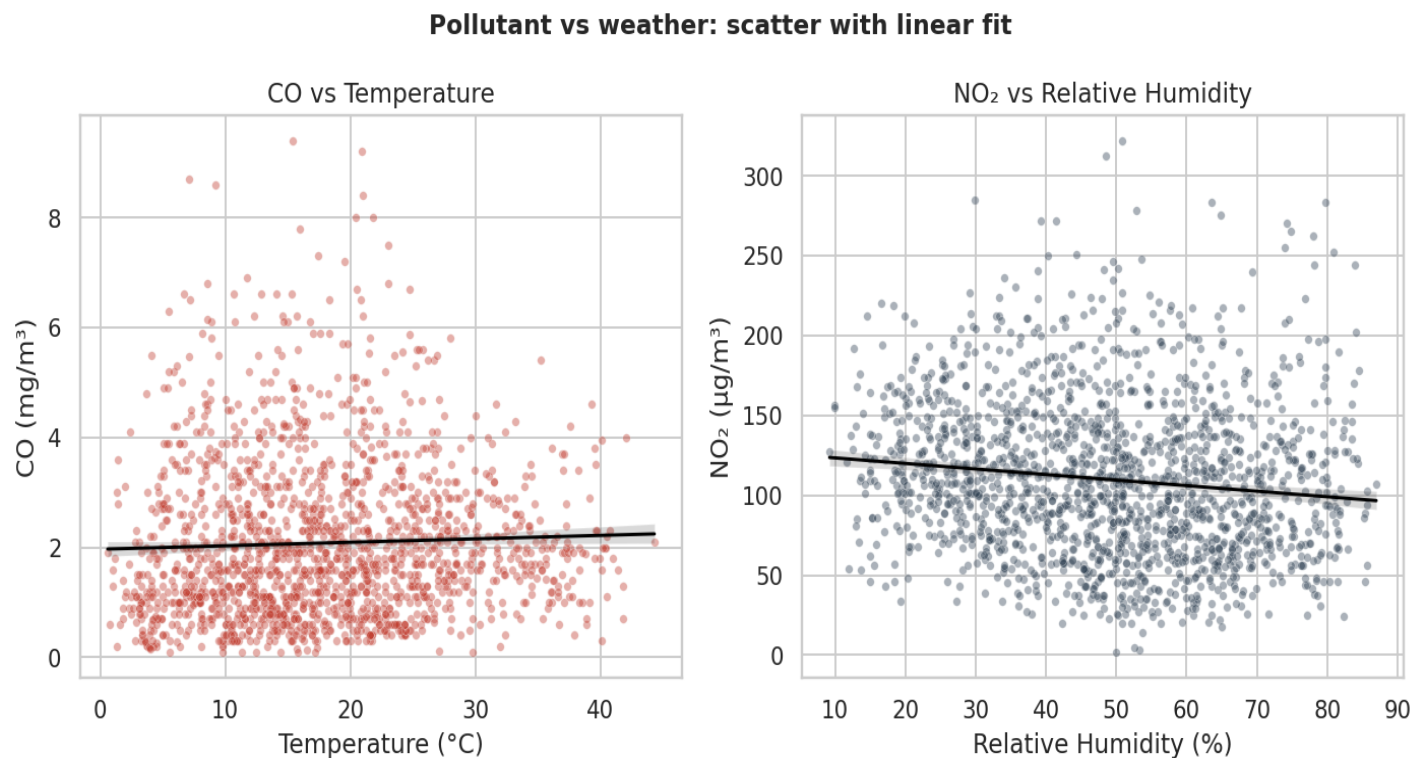


Figure 6 — Scatter plots with linear fit (random sample of 2,000 hours).

4. Predictive Modelling

As an optional analytical step, two regression models were trained to predict hourly benzene (C₆H₆) from co-located readings of CO, NO_x, NO₂, weather, and time-of-day features. The dataset was split chronologically — first 80 % for training, final 20 % for testing — to avoid temporal leakage.

Model	MAE (µg/m³)	R²
Linear regression	2.26	0.77
Random forest (120 trees)	1.76	0.77

Both models achieve $R^2 \approx 0.77$; however, the Random Forest reduces mean absolute error from 2.26 to 1.76 µg/m³, an improvement of approximately 22%, indicating better absolute prediction accuracy on the held-out period. Predicted vs actual benzene over the held-out period (Figure 7) tracks the diurnal and weekly cycles closely; the model underestimates the highest spikes, a typical regression-to-the-mean behaviour.

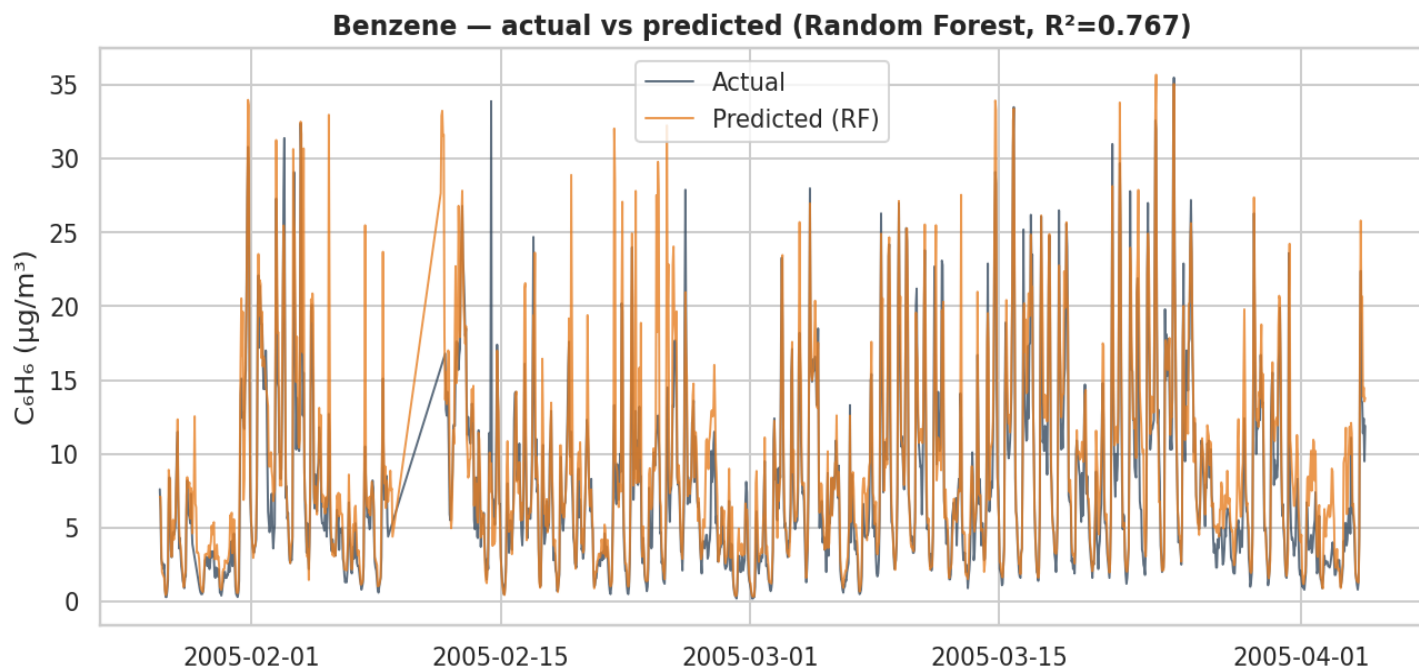


Figure 7 — Random-Forest benzene predictions on the held-out 20 % of the timeline.

Feature importance (Figure 8) is dominated by CO, consistent with the strong CO–benzene correlation ($r = 0.91$) observed earlier. NO_x and time-of-day features make moderate contributions, while weather variables have relatively low individual importance in the model.

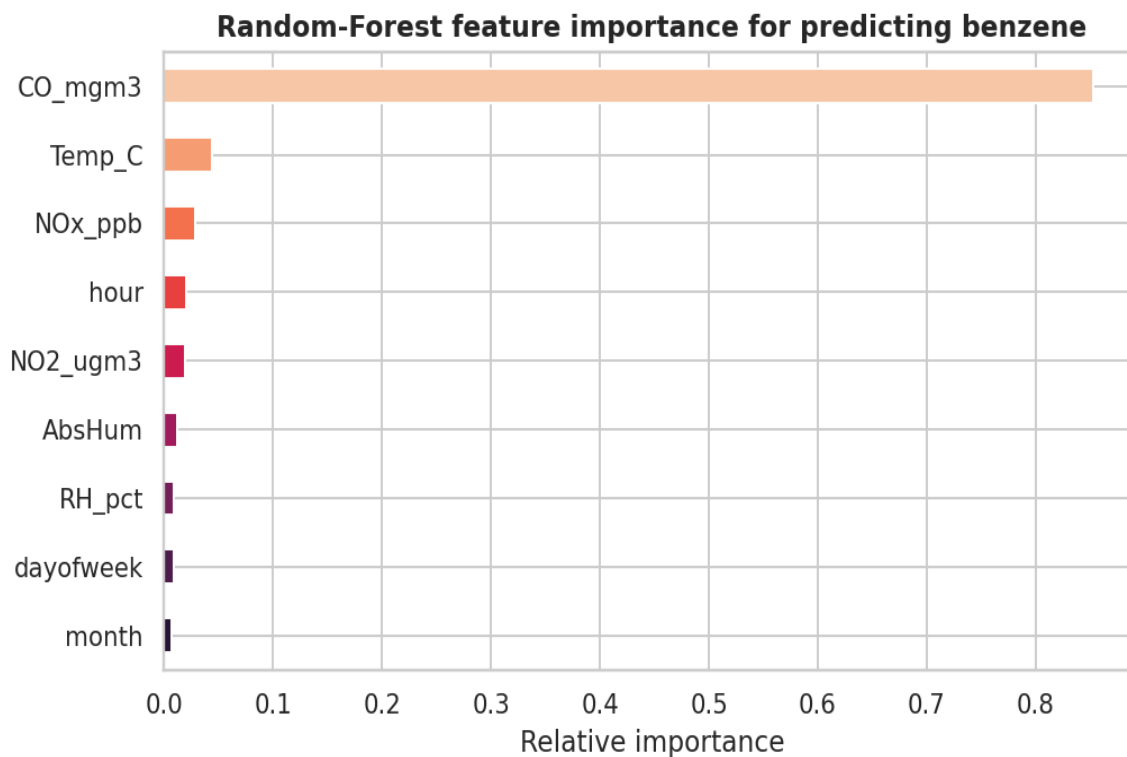


Figure 8 — Random-Forest feature importance for the benzene model.

5. Discussion and Implications

5.1 Public-health implications

Benzene is a carcinogenic air pollutant for which WHO states that no safe level of exposure can be recommended. WHO estimates that lifetime exposure to $1.7 \mu\text{g}/\text{m}^3$ is associated with an excess lifetime leukaemia risk of 1 in 100,000. The mean benzene concentration observed in this dataset ($10.1 \mu\text{g}/\text{m}^3$) is also approximately twice the EU calendar-year limit value of $5 \mu\text{g}/\text{m}^3$. These comparisons underscore the public-health relevance of the concentrations observed, although this 13-month single-site dataset should not be interpreted as an individual exposure or lifetime-risk assessment.

NO_2 is associated with respiratory inflammation, asthma exacerbation and reduced lung function. Hourly NO_2 concentrations in the dataset reach a maximum of $340 \mu\text{g}/\text{m}^3$, with some observations exceeding $200 \mu\text{g}/\text{m}^3$. A formal exceedance analysis would be required to determine the frequency and regulatory significance of these high-concentration episodes.

5.2 Policy implications

- The 27% weekend reduction in NO_x highlights the potential importance of traffic activity in urban pollution patterns and provides a strong rationale for investigating traffic-management interventions.
- The winter peaks in NO_x and NO_2 highlight the importance of seasonal conditions in air-quality management, including changes in emissions and atmospheric dispersion during colder months.
- The strong CO–benzene relationship suggests that interventions targeting shared combustion sources could potentially reduce multiple pollutants simultaneously.

5.3 Limitations

- Single-site dataset: results are not directly transferable to other urban contexts.
- 13-month coverage prevents proper inter-annual trend or policy-impact assessment.
- NMHC was discarded due to >90 % missingness, leaving total non-methane hydrocarbons unmeasured.
- Correlation does not equal causation; weather effects are partially confounded with seasonal traffic patterns.

5.4 Recommended next steps

- Compare against contemporaneous regulatory monitoring stations to validate sensor accuracy.
- Extend the model to ozone (O_3) using the PT08.S5 sensor, which was omitted here.
- Apply a state-space or seasonal ARIMA approach for formal trend extraction.
- Conduct a formal NO_2 exceedance analysis using the applicable regulatory threshold, averaging period and permitted exceedance frequency.

Appendix · Summary Statistics

Variable	n	Mean	Std	Min	Median	Max
CO (mg/m^3)	8,176	2.13	1.46	0.10	1.80	11.90
C_6H_6 ($\mu\text{g}/\text{m}^3$)	9,117	10.14	7.51	0.10	8.30	63.70
NO_x (ppb)	8,295	240.85	209.99	2	175	1479
NO_2 ($\mu\text{g}/\text{m}^3$)	8,293	111.03	48.28	2	107	340
Temp ($^{\circ}\text{C}$)	9,117	18.31	8.83	-1.9	17.8	44.6
RH (%)	9,117	49.17	17.30	9.2	49.5	88.7